

**CELEBAL TECHNOLOGIES**

Data Science Internship

**MAJOR PROJECT REPORT**

**Satellite Land-Use Classification**

**&**

**Temporal Change Detection**

**Using Deep Learning**

**Submitted By**

**Akshit Sharma**

B.Tech Computer Science Engineering

(Artificial Intelligence)

Poornima College of Engineering

Celebal Technologies

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## Abstract

This project presents an end-to-end deep learning system for land-use classification and temporal change detection using satellite imagery. A fine-tuned ResNet18 model was developed to classify images from the EuroSAT dataset into ten land-use categories. The trained model was further utilized as a feature extractor to generate image embeddings for temporal change detection. Cosine similarity between image embeddings was used to determine whether significant land-use change had occurred, followed by heatmap visualization for interpretation. A Streamlit-based web application was developed to provide an interactive interface for classification and change detection. The project demonstrates a practical workflow for integrating transfer learning, feature embeddings, and deployment into a complete remote sensing application.

## 2. Problem Statement

Manual monitoring of land-use changes from satellite imagery is time-consuming, resource-intensive, and difficult to scale. Conventional approaches require extensive human interpretation, making continuous monitoring challenging. This project addresses these limitations by developing an automated deep learning system capable of classifying land-use categories and identifying temporal changes between satellite images through feature embedding similarity and an interactive web application.

## 3. Objectives

- Develop a deep learning model for land-use classification using the EuroSAT dataset.
- Apply transfer learning with ResNet18 to improve classification performance.
- Extract deep feature embeddings for temporal change detection.
- Detect land-use changes using cosine similarity and an adaptive threshold.
- Develop and deploy an interactive Streamlit application for real-time inference.

## 4. Overall Workflow

The proposed system follows an end-to-end workflow for land-use classification and temporal change detection using satellite imagery. EuroSAT images are preprocessed and passed to a fine-tuned ResNet18 model. The same model performs land-use classification and extracts deep feature embeddings. Cosine similarity between embeddings is used to detect significant land-use changes, and the final results are visualized through a Streamlit application.

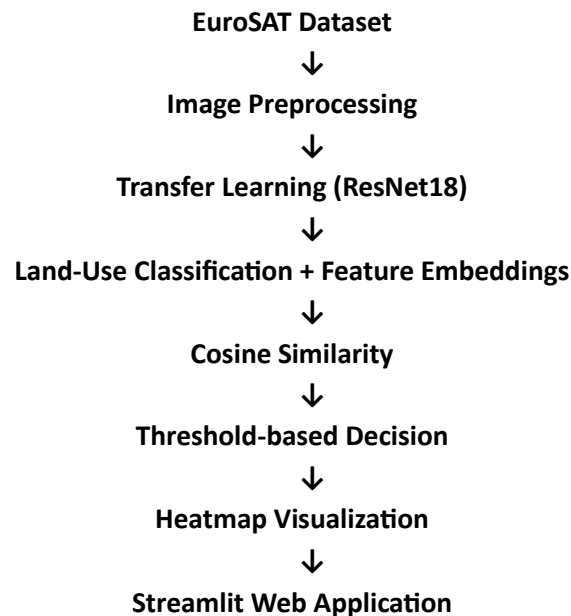


Figure 1. Overall workflow of the proposed system.

## 5. Dataset Description

The EuroSAT dataset was used to train and evaluate the proposed model. It contains Sentinel-2 satellite images belonging to ten land-use classes. Each RGB image has a spatial resolution of  $64 \times 64$  pixels and is organized into class-wise directories for supervised learning.

| Property     | Value          |
|--------------|----------------|
| Dataset      | EuroSAT        |
| Total Images | 27,000         |
| Classes      | 10             |
| Resolution   | $64 \times 64$ |
| Satellite    | Sentinel-2     |

## 6. Model Development

The land-use classification model was developed using transfer learning with a pre-trained ResNet18 backbone. A baseline CNN was initially implemented to establish the training pipeline and understand the EuroSAT dataset. The final implementation adopted ResNet18 because it provided better feature extraction capability and faster convergence while remaining suitable for deployment.

6.1 Two-Phase Fine-Tuning Strategy

Training was performed in two stages to efficiently adapt the pre-trained network to satellite imagery.

| Phase   | Description                                                                    |
|---------|--------------------------------------------------------------------------------|
| Phase 1 | Freeze the ResNet18 backbone and train only the classifier head.               |
| Phase 2 | Unfreeze the final layers and fine-tune the model for better feature learning. |

6.2 Model Configuration

| Parameter  | Value     |
|------------|-----------|
| Framework  | PyTorch   |
| Backbone   | ResNet18  |
| Dataset    | EuroSAT   |
| Input Size | 224 × 224 |
| Classes    | 10        |

6.3 Training Pipeline

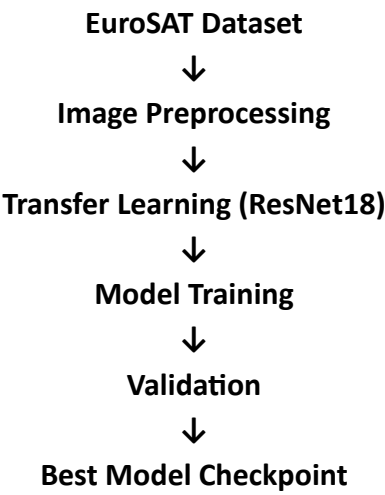


Figure 2. Training pipeline adopted for the proposed ResNet18 model.

## 7. Temporal Change Detection

Beyond land-use classification, the proposed system performs temporal change detection using deep feature embeddings extracted from the fine-tuned ResNet18 model. Instead of comparing raw pixel values, cosine similarity is computed between the embeddings of two satellite images. The similarity score is compared with a predefined threshold to determine whether significant land-use change has occurred.

### 7.1 Feature Embedding Extraction

The fine-tuned ResNet18 model acts as both a classifier and a feature extractor. For each input image, a 512-dimensional embedding is generated to capture semantic information required for change analysis.

### 7.2 Similarity Threshold

| Parameter          | Value             |
|--------------------|-------------------|
| Embedding Size     | 512               |
| Similarity Metric  | Cosine Similarity |
| Decision Threshold | 0.6345            |

### 7.3 Change Detection Workflow

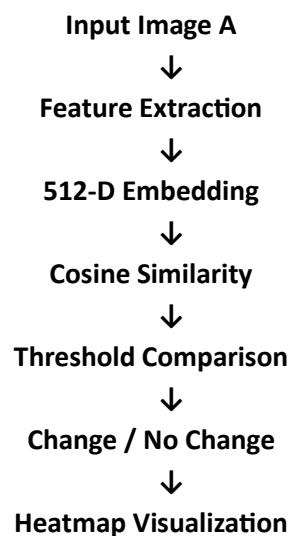
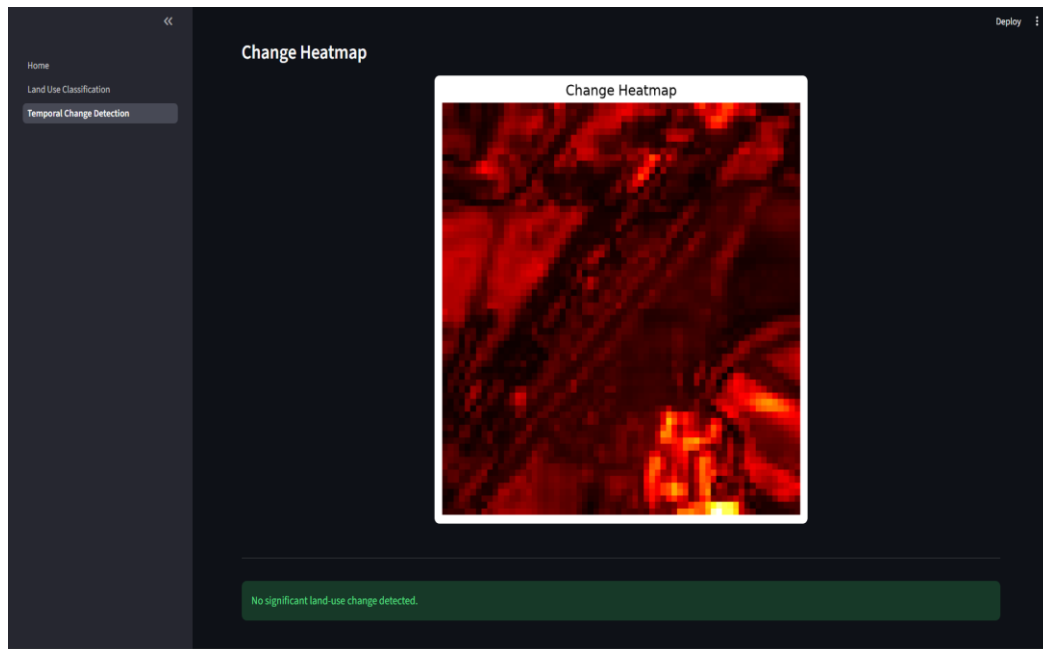


Figure 4. Workflow adopted for temporal change detection.

### 7.4 Heatmap Visualization

The generated heatmap highlights the regions with maximum feature variation between the two temporal satellite images. The visualization improves the interpretability of the similarity-based change detection process.



.Figure 5 Heatmap generated for temporal land-use change visualization.

## 8. Experimental Results

The proposed system was evaluated on the EuroSAT dataset after fine-tuning the ResNet18 model. The developed Streamlit application was used to validate land-use classification and temporal change detection using unseen satellite images. The results demonstrate reliable classification performance together with an intuitive visualization of detected changes.

### 8.1 Performance Summary

| Metric                  | Value   |
|-------------------------|---------|
| Classification Accuracy | 96.48%  |
| Dataset                 | EuroSAT |
| Land-use Classes        | 10      |
| Embedding Dimension     | 512     |
| Decision Threshold      | 0.6345  |

### 8.2 Classification Results

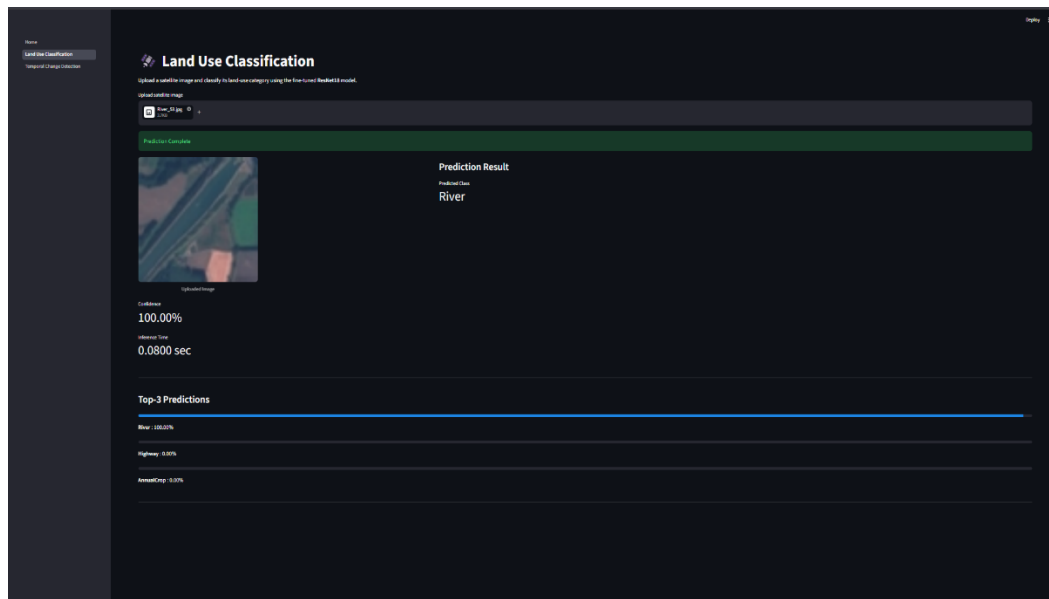
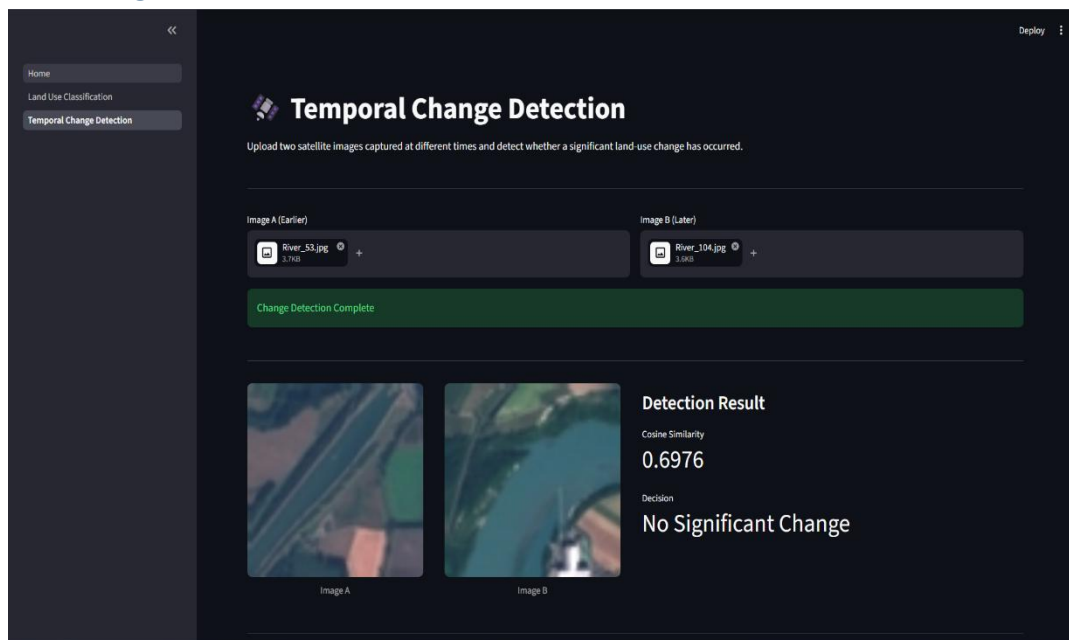


Figure 6. Land-use classification result generated by the Streamlit application.

The uploaded satellite image was successfully classified using the fine-tuned ResNet18 model. The application displays the predicted land-use class, confidence score, inference time, and the top three predictions, demonstrating the effectiveness of the proposed classification pipeline.

### 8.3 Change Detection Results



The cosine similarity between the two satellite images was 0.6976, which is greater than the selected decision threshold of 0.6345. Therefore, the model correctly identified that no significant land-use change had occurred between the input images.

## 9. Streamlit Dashboard & Deployment

The proposed deep learning system was successfully deployed as an interactive Streamlit web application for land-use classification and temporal change detection. The dashboard enables users to upload satellite images, perform real-time inference, visualize prediction confidence, compare temporal images using cosine similarity, and generate change heatmaps.

### 9.1 Dashboard Features

- Interactive Streamlit interface
- Land-use classification using ResNet18
- Top-3 predictions with confidence
- Temporal change detection
- Cosine similarity computation
- Heatmap visualization
- Fast real-time inference

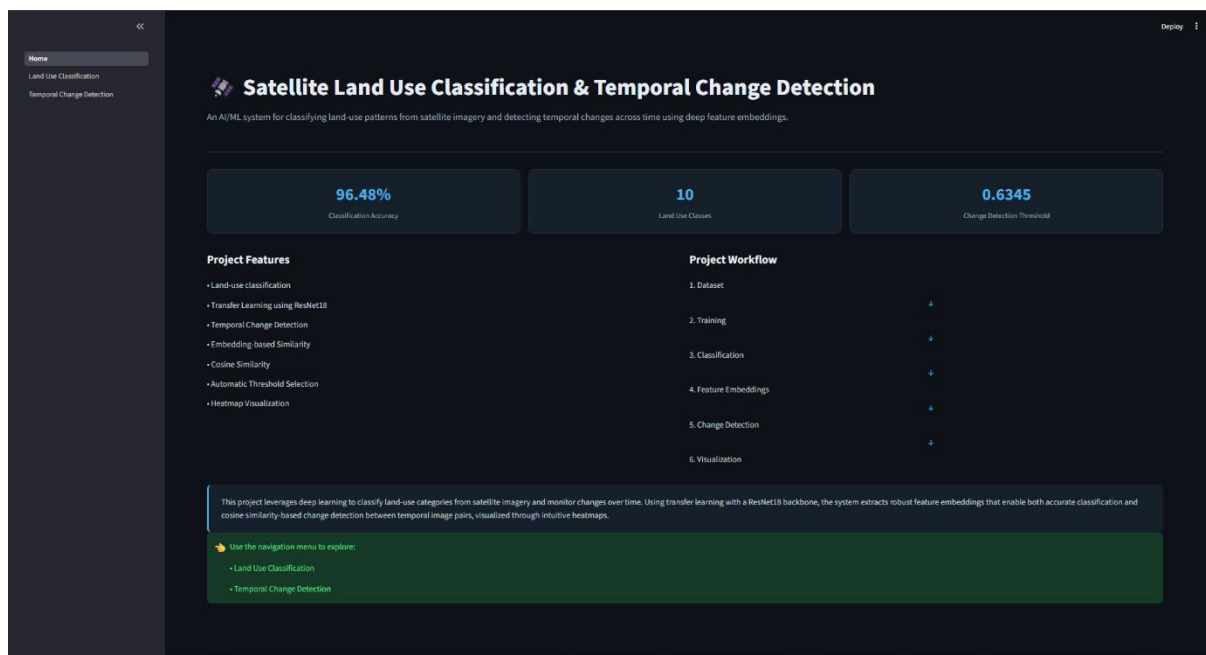


Figure 8. Streamlit dashboard developed for the proposed system.

## 10. Challenges Faced

Fine-tuning the ResNet18 model required careful optimization to achieve stable convergence while avoiding overfitting. Selecting an appropriate cosine similarity threshold was another important challenge because it directly influenced change detection performance. Integrating classification and temporal change detection into a single Streamlit application required additional implementation effort.

## 11. Limitations

- Model evaluated primarily on the EuroSAT dataset.
- Generalization to other datasets requires further validation.
- Change detection is based on feature embedding similarity.
- Performance may vary across sensors and image resolutions.

## 12. Conclusion

This project successfully developed an end-to-end deep learning system for satellite land-use classification and temporal change detection. Transfer learning with ResNet18 achieved reliable performance, while embedding-based cosine similarity enabled effective temporal analysis. The Streamlit application demonstrated the practical deployment of the proposed framework.

## 13. Future Work

- Evaluate on additional satellite datasets.
- Explore Vision Transformer architectures.
- Implement pixel-level change segmentation.
- Deploy as a cloud-hosted application.

## References

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